

Basics of Shared-Memory Concurrency

Read-Modify-Write (RMW):

- Holds some value x
- Has a method `getAndMumble()` such that: Returns object's prior value x . Replaces x with $mumble(x)$ for some function $mumble$. And it does so **atomically**, in a mutually exclusive way.

Weak RMW: enables synchronisation between **two** threads. (E.g. exchange – `xchg`, FetchAndAdd-FAA)

Strong RMW: enables synchronisation between an **arbitrary** number of threads. (E.g. CompareAndSet-CAS)